

NANYANG TECHNOLOGICAL UNIVERSITY
School of Social Sciences
HE3002 Macroeconomics III

LECTURE 5
Exchange Rate Regimes

READINGS

1. Olivier Blanchard *Macroeconomics*, 8th Edition (2021), Prentice Hall, Chapter 20.

THE MEDIUM RUN

1. Under flexible exchange rates, a country seeking real depreciation to reduce its trade deficit or recover from a recession could achieve this through expansionary monetary policy, leading to lower interest rates and a depreciated exchange rate.
2. Under fixed exchange rates, a country lost both instruments: the nominal exchange rate couldn't be adjusted, and the UIP condition required the domestic interest rate to match the foreign rate, preventing interest rate adjustments.
3. This makes a flexible exchange rate regime more attractive than a fixed exchange rate regime.
4. Why should a country give up two macroeconomic instruments?
5. In the medium run, when prices adjust, the difference between the two regimes fades away.
6. In the medium run, the economy reaches the same real exchange rate and the same level of output, whether it operates under fixed or flexible exchange rates.
7. Recall the definition of the real exchange rate:

$$\epsilon = \frac{EP}{P^*} \quad (1)$$

8. There are two ways in which the real exchange rate can adjust:
 - (a) Through a change in nominal exchange rate, E . By definition, this can be done only under flexible exchange rates. And if we assume that P , and P^* , do not change in the short run, the only way to adjust the real exchange rate in the short run is through change in E .
 - (b) Through a change in P , relative to P^* in the medium run. This option is available even to a country operating under a fixed (nominal) exchange rate regime. The adjustment takes place through the price level rather than through the nominal exchange rate.

The IS Relation under Fixed Exchange Rates

1. In an open economy with fixed exchange rates, we can write the IS relation as:

$$Y = Y \left(\frac{\bar{E}P}{P^*}, G, T, i^* - \pi^e, Y^* \right) \quad (2)$$

This is because

- (a) The real interest rate, r is equal to the nominal interest rate, i minus expected inflation, π^e :

$$r = i - \pi^e$$

- (b) The real exchange rate, ϵ

$$\epsilon = \frac{EP}{P^*}$$

- (c) Under fixed exchange rate,

$$E = \bar{E}$$

- (d) Under fixed exchange rates and perfect capital mobility,

$$i = i^*$$

2. Demand, and hence output, depend:

- (a) negatively on the real exchange rate.
- (b) positively on government spending and negatively on taxes.
- (c) negatively on the domestic real interest rate.
- (d) positively on foreign output through the effect on exports.

Equilibrium in the Short and the Medium Run

1. Consider an economy where the real exchange rate is too high. As a result, the trade balance is in deficit, $NX < 0$ and output is below potential, $Y < Y_n$.
2. Under a flexible exchange rate regime,
 - (a) The central bank could decrease the interest rate, induce a nominal depreciation.
 - (b) Given the domestic and foreign price levels, which is fixed in the short run, the nominal depreciation would imply a real depreciation, an improve in trade balance and an increase in output.
3. Under a fixed exchange rate regime,

- (a) The central bank cannot move the domestic interest rate. Thus, in the short run, the trade deficit remains, and the country remains in recession.
- (b) In the medium run, price can adjust.
- (c) The Phillips curve relation:

$$\pi - \pi^e = (\alpha/L)(Y - Y_n) \quad (3)$$

When $Y > Y_n$, then $\pi > \pi^e$. When $Y < Y_n$, then $\pi < \pi^e$.

- (d) Suppose if expected inflation is assumed to be constant, then, $\pi^e = \bar{\pi}$. This means that:

$$\pi - \bar{\pi} = (\alpha/L)(Y - Y_n) \quad (4)$$

- (e) Suppose that if both economies are at potential, inflation rates were the same, relative price levels would remain constant, and so would the real exchange rate. i.e. $\pi = \pi^* = \bar{\pi}$.
- (f) As we assume that $Y < Y_n$, equation (4) implies that domestic inflation is lower than it would be if $Y = Y_n$, and thus lower than foreign inflation.
- (g) This means that the domestic price level increases more slowly than the foreign price level. Given the nominal exchange rate, which is fixed, the real exchange rate decreases.
- (h) As a result, net exports increases over time, and so does output.

4. To summarize,

- (a) In the short run, a fixed nominal exchange rate implies a fixed real exchange rate.
- (b) In the medium run, the real exchange rate can adjust even if nominal exchange rate is fixed. This adjustment is achieved through movements in relative price levels over time.

EXCHANGE RATE CRISES UNDER FIXED EXCHANGE RATES

1. Suppose a country has chosen to operate under a fixed exchange rate.
2. Suppose financial investors begin to anticipate an exchange rate adjustment, either through devaluation or a shift to a flexible exchange rate regime with depreciation.
 - (a) The real exchange rate may be too high. Or, the domestic currency may be over-valued. In this case, a real depreciation is called for. Although this could be achieved in the medium run without a devaluation, financial investors might conclude that the government will take the quickest way out and devalue. Such an overvaluation is likely to happen in a country that pegs its nominal exchange rate to the currency of a country with lower inflation. Higher relative inflation implies a steadily increasing price of domestic goods relative to foreign

goods, a steady real appreciation, and so a steady worsening of the trade position. As time passes, the need for an adjustment of the real exchange rate increases, and financial investors become more and more nervous. They start thinking that a devaluation might be coming.

(b) Internal conditions may demand a lower domestic interest rate, which is not possible under fixed exchange rates. However, it can be achieved by switching to a flexible exchange rate regime. If the country allows the exchange rate to float and lowers its domestic interest rate, this will lead to a nominal depreciation.

3. As soon as financial markets believe a devaluation may be coming, maintaining the exchange rate requires an increase, often a large one, in the domestic interest rate.
4. To see this, return to the interest parity condition:

$$i_t = i_t^* - \frac{(E_{t+1}^e - E_t)}{E_t} \quad (5)$$

5. We interpreted this equation as a relation between the **one-year** domestic and foreign nominal interest rates, the current exchange rate and the expected exchange rate **a year**. But the choice of one year as the period was arbitrary.
6. The relation holds over short periods such as a day, week, or month. If financial markets expect the exchange rate to drop by 2% in a month, they will only hold domestic bonds if the one-month domestic interest rate exceeds the foreign interest rate by 2%. Expressed annually, this means the domestic interest rate must be 24% higher than the foreign rate ($2\% \times 12 = 24\%$).
7. Under fixed exchange rates, the current exchange rate, E_t , is set at some level, say $E_t = \bar{E}$.
8. If markets expect that the parity will be maintained over the period, then $E_{t+1}^e = \bar{E}$, and the UIP condition simply states that the domestic and the foreign interest rates must be equal.
9. Suppose, however, that participants in financial markets start anticipating a devaluation, a decrease in the exchange rate.
10. Suppose they believe that, over the coming month, there is a 75% chance the parity will be maintained and a 25% chance there will be a 20% devaluation.
11. The term $(E_{t+1}^e - E_t)/E_t$ which we assumed equal to 0 earlier, now equals:

$$0.75 \times 0\% + 0.25 \times 20\% = 5\%$$

12. That is, a 75% chance of no change plus a 25% chance of a devaluation of 20%.

13. This implies that, if the central bank wants to maintain the existing parity, it must now set a **monthly** interest rate 5% higher than before – 60% higher **annually** ($12 \text{ months} \times 5\% \text{ per month}$) is the interest differential needed to convince investors to hold domestic bonds rather than foreign bonds.
14. Any smaller interest differential, and investors will not want to hold domestic bonds.
15. What choices, then, do the government and the central bank have?
 - (a) The government and central bank can attempt to reassure markets by asserting they have no plans to devalue. Prime ministers may appear on TV to reaffirm their strong commitment to maintaining the current exchange rate parity.
 - (b) The central bank may raise interest rates, but not sufficiently to offset the perceived risk of devaluation. While domestic interest rates are high, they still fail to fully compensate investors for the risk. As a result, large financial outflows occur, as investors shift from domestic to foreign bonds. They sell domestic bonds for domestic currency, then exchange it for foreign currency to purchase foreign bonds.
 - i. **No intervention:** The large sales of domestic currency for foreign currency would lead to a depreciation.
 - ii. **With intervention:** The central bank must therefore stand ready to buy domestic currency and sell foreign currency at the current exchange rate. In doing so, it often loses most of its reserves of foreign currency.
 - (c) Eventually, the choice for the central bank becomes either to increase the interest rate enough to satisfy Equation (5), or to validate the market's expectations and devalue.
 - (d) Setting a very high short-term domestic interest rate can have a devastating effect on demand and on output – no firm wants to invest and no consumer wants to borrow when interest rates are very high.
 - (e) This course of action makes sense only if (1) the perceived probability of a devaluation is small, so the interest rate does not have to be too high, and (2) the government believes markets will soon become convinced that no devaluation is coming, allowing domestic interest rates to decrease.
 - (f) Otherwise, the only option is to devalue.
16. To summarise, expectations that a devaluation may be coming can trigger an exchange rate crisis. Faced with such expectations, the government has two options:
 - (a) Give in and devalue.
 - (b) Fight and maintain the parity, at the cost of very high interest rates and a potential recession.

EXCHANGE RATE MOVEMENTS UNDER FLEXIBLE EXCHANGE RATE

1. In the model we developed, there was a simple relation between the interest rate and the exchange rate: the lower the interest rate, the lower the exchange rate.
2. This implied that a country that wanted to maintain a stable exchange rate just had to maintain its interest rate close to the foreign interest rate.
3. A country that wanted to achieve a given depreciation just had to decrease its interest rate by the right amount.
4. In reality, exchange rates often move even in the absence of movements in interest rates. Furthermore, the size of the effect of a given change in the interest rate on the exchange rate is hard to predict.
5. This makes it much harder for monetary policy to achieve its desired outcome.
6. To see why things are more complicated, we must return once again to the interest parity condition:

$$(1 + i_t) = (1 + i_t^*) \left(\frac{E_t}{E_{t+1}^e} \right) \quad (6)$$

7. Arranging this gives:

$$E_t = \frac{1 + i_t}{1 + i_t^*} E_{t+1}^e \quad (7)$$

8. Think of the time period (from t to $t + 1$) as one year.
9. The exchange rate this year depends on the one-year domestic interest rate, the one-year foreign interest rate and the exchange rate expected for next year.
10. We assumed that the expected exchange rate next year, E_{t+1}^e was constant, but that was a simplification.
11. The exchange rate expected one year hence is not constant.
12. Using Equation (7), but now for next year, we can write Equation (7) for year $t + 1$ as:

$$E_{t+1} = \frac{1 + i_{t+1}}{1 + i_{t+1}^*} E_{t+2}^e \quad (8)$$

13. The exchange rate in year $t + 1$ depends on the domestic interest rate and the foreign interest rate for year $t + 1$, as well as on the expected future exchange rate in year $t + 2$. So, the expectation of the exchange rate in year $t + 1$, held as of year t , is given by:

$$E_{t+1}^e = \frac{1 + i_{t+1}^e}{1 + i_{t+1}^{*e}} E_{t+2}^e \quad (9)$$

14. Replacing E_{t+1}^e with the expression above gives:

$$E_t = \frac{(1 + i_t)(1 + i_{t+1}^e)}{(1 + i_t^*)(1 + i_{t+1}^{*e})} E_{t+2}^e \quad (10)$$

15. The current exchange rate depends on this year's domestic and foreign interest rates, on next year's expected domestic and foreign interest rates, and on the expected exchange rate two years from now. Continuing to solve forward in time in the same way we get:

$$E_t = \frac{(1 + i_t)(1 + i_{t+1}^e) \dots (1 + i_{t+n}^e)}{(1 + i_t^*)(1 + i_{t+1}^{*e}) \dots (1 + i_{t+n}^{*e})} E_{t+n+1}^e \quad (11)$$

16. Suppose we take n to be large, say 10 years, this relation tells us that the current exchange rate depends on two sets of factors:

- (a) Current and expected domestic and foreign interest rates for each year over the next ten years.
- (b) The expected exchange rate 10 years from now.

17. Equation (11) makes the three points we want to make here:

- (a) The level of today's exchange rate will move one-for-one with the future expected exchange rate.
- (b) Today's exchange rate will move when future expected interest rates move in either country.
- (c) Because today's exchange rate moves with any change in expectations, the exchange rate will be volatile.